# William Abbruzzese

Portland, Oregon | w_abbruzzese2024@coloradocollege.edu | 503 913 1197 | willabbruzzese.github.io

## Education

**Colorado College**, Bachelor's in Physics — May 2028

**GPA:** 3.96/4.0, Dean's List Fall 2024–Present
**Relevant Coursework:** Computational Physics, Mechanics (orbital), Observational Astronomy, Electronics, Techniques of Experimental Physics, Modern Physics, Applied Linear Algebra

## Research & Engineering Projects

**Autonomous UAV GNC Research** | Colorado College | *Principal Investigator & Lead Engineer* — April 2026 – Present

- Awarded a $5,000 institutional grant to design and build a search & rescue drone from the frame up, leading a team of 3 engineers and partnering with Multnomah County Sheriff's Office Search & Rescue.
- Wired and integrated avionics, GPS, telemetry, power, and propulsion electronics; built a custom MAVLink ground control station for interactive search-area planning and live telemetry.
- Designed terrain-optimized guidance algorithms reducing search time by 9%, integrating Bayesian search theory and Koopman kinematic models to dynamically update target zones using Probability of Detection metrics.
- Developed a three-tier natural language command parser translating voice into flight trajectories; validated via 25 autonomous flight tests with a 96% command-execution success rate.

**Optical Tracking of Artemis II Orion Spacecraft** | Colorado College — April 2026

- Generated ephemerides through NASA's Horizons System and integrated them into TheSkyX to acquire and track Orion during its trans-Earth coast, resolving a non-cataloged target at roughly lunar distance.
- Captured a three-frame sequence resolving the capsule's motion against a fixed star field.

**Rocket Avionics & Control Systems** | Independent Project — June 2025 – Aug 2025

- Developed and deployed an Arduino-based flight data acquisition payload using an onboard IMU and barometric altimeter to track real-time rocket trajectories.
- Engineered a fully analog PID controller from discrete components, simulating in LTSpice and fabricating a custom PCB to validate 2D state-space stabilization on a self-balancing platform.
- Analyzed telemetry from 20+ launch trajectories in MATLAB and Python; designed a Kalman filter to reject high-frequency structural noise and reconstruct altitude and velocity profiles.

**Micro-Wind Turbine Electromechanical Design** | Independent Volunteer Project — June 2023 – Aug 2023

- Designed and manufactured a functional small-scale wind turbine from salvaged electromechanical components, implementing circuit fabrication and power distribution to deliver reliable lighting to a rural facility in Nicaragua lacking electrical infrastructure.

## Leadership & Professional Experience

**Team Leader** | Multnomah County Search and Rescue – Portland, OR — Sept 2021 – Aug 2024

- Directed field operations for cross-functional teams of 10–15 individuals in high-pressure environments, emphasizing safety, communication, and rapid problem-solving.
- Logged 1,000+ operational hours as a 911 wilderness first responder locating and assisting missing or injured individuals in remote terrain throughout Oregon.
- Coordinated training and field operations as highest-ranked Team Leader, strengthening leadership, stakeholder coordination, and high-reliability teamwork.

**Outdoor Gear Technician** | Ahlberg Gear House – Colorado Springs, CO — March 2025 – Present

- Repair and calibrate technical outdoor equipment to safety-critical tolerances; advise customers on gear selection and trip planning, and manage inventory systems tracking thousands of items.

## Technical Skills

- **Programming & Software:** Python, MATLAB, Mathematica, C++, SolidWorks, LTSpice, Altium, TheSkyX.
- **Embedded & Robotics:** ArduPilot, PX4, MAVLink, MAVProxy, Arduino, QGroundControl
- **Languages:** Italian, Spanish (Oregon State Seal of Biliteracy).